

Machine learning models of antibody-excipient preferential interactions for use in computational formulation design

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Abstract

Preferential interactions of formulation excipients govern their impact on the stability properties of proteins in solution. The ability to predict these interactions without the need to perform experiments would enable formulation design to begin early in the development of a new antibody therapeutic. With that in mind, we developed a feature set to numerically describe local regions of an antibody’s surface for use in machine learning applications. Then, we used these features to train machine learning models for local antibody-excipient preferential interactions for the excipients sorbitol, sucrose, trehalose, proline, arginine.HCl, and NaCl. Our models had accuracies of up to about 85%. We also used linear (elastic net) models to quantify the contribution of antibody surface features to the preferential interaction coefficients, finding that the carbohydrates and proline tend to have similar important features, while the interactions of arginine.HCl and NaCl are governed by charge features. We present several case studies demonstrating how these machine learning models could be used to predict experimental aggregation and viscosity behavior in solution. Finally, we propose an approach to computational formulation design wherein a panel of may be considered while designing an antibody sequence.

Abbreviations

EN,SVM,mAb,RMSE,Fv,MD,SAP,SCM

Introduction

Monoclonal antibodies (mAbs) are a rapidly growing class of therapeutic proteins that are frequently formulated at high concentrations.^{1,2} At high concentrations, physical instabilities, such as aggregation and viscosity, often must be addressed.^{3,4} These instabilities can be mitigated by mutating the antibody so that it is more stable or by a trial and error selection of formulation excipients, but often these do not work.¹ One of the reasons is that, even though

some general trends in the impacts of certain classes of excipients on mAb behavior have been identified, formulation design generally requires an experimental screening approach, which is costly in terms of time and material.⁴

There has been significant work in understanding how the antibody sequence and structure tends to impact aggregation and viscosity behaviors.⁵⁻⁸ However, these studies rarely consider the impact of excipients, possibly because the mechanisms of antibody-excipient interactions are generally not well understood, and there can be significant variations in the impact of the same excipient on different mAbs.⁹⁻¹¹ Timasheff and colleagues pioneered the study of protein-excipient preferential interactions, examining how preferential hydration of globular proteins affected stability.¹²⁻¹⁵ More recently, we have measured the preferential interactions of a variety of excipients with three different mAbs.^{16,17}

However, these experimental approaches are limited to studying the global or average interactions of excipients with entire proteins, but do not measure the local interactions. Antibody aggregation and viscosity behaviors are thought to be driven by local antibody-antibody interactions, for example with particular interactions of hydrophobic patches leading to aggregation events.⁸ Therefore, the local interactions of excipients with this particular patch may directly affect the aggregation behavior. Even if an excipient is globally (on average) excluded from the antibody, it may be locally included near a hydrophobic surface patch and thus may have the potential to disrupt aggregation.

While it is possible to examine certain local behaviors experimentally, such as by using HD-exchange to understand how local mAb flexibility changes in the presence of an excipient,¹⁸ molecular-level resolution of these interactions is best accessed via simulation. HD-exchange is sensitive at the peptide level,¹⁸ while simulations provide residue-level resolution. Recently, Tilegenova et al. studied the interactions of mono- and bi-specific antibodies with arginine.HCl (Arg.HCl) via experiment and simulation, identifying point mutations that reduced viscosity.¹⁹ Various other computational approaches have been used to study protein-excipient interactions,²⁰⁻²² including our previous studies on antibody-excipient in-

teractions using molecular dynamics (MD) simulations.^{9,23} However, simulations are often opaque, as they do not directly yield a numerical relationship between chemical surface properties and excipient interactions. Instead, they require lengthy manual analysis of the results to draw conclusions about protein-excipient interaction mechanisms.^{19,22,23}

Machine learning approaches can be used to analyze simulation results to look for trends in interactions. While machine learning approaches have frequently been used for other antibody applications, including antibody structure prediction²⁴ and antigen binding,^{25–27} they have not yet been applied to model antibody-excipient interactions. Linear models will yield a direct numerical relationship between antibody surface properties and the local excipient concentration, although likely with less accuracy than nonlinear models, but it is more difficult to extract physical insight from nonlinear models. If reasonably high accuracy is achieved with a nonlinear model, antibody-excipient interactions could be predicted in a matter of seconds instead of days. Rapid prediction of antibody-excipient interactions, as well as an understanding of the impact of these interactions on physical stability of antibodies, is desired for early stage formulation design. This would reduce the number of experiments required, reducing the overall cost of formulation development.

Using machine learning models requires numerically representing the antibody surface. Previously, various groups have worked to identify features that might be important in describing the proteins, including representations of amino acids,²⁸ molecular curvature,²⁹ and 3D Zernike descriptors for mAbs.³⁰ However, thus far, no systematic framework for representing the chemical properties of local regions of a mAb’s surface has been developed. Additionally, to the best of our knowledge, no systematic studies of how antibody surface properties affect the interactions with a large panel of excipients exist beyond our previous work.^{9,23}

Here, we use machine learning algorithms to connect antibody surface features to local preferential interactions with a variety of excipients. First, a feature set for numerically describing local surface regions of mAbs is developed. Machine learning models are trained

using these features to predict local mAb-excipient preferential interactions as calculated via MD simulation. We demonstrate the use of this approach on formulation excipients sorbitol, sucrose, trehalose, proline, Arg.HCl, and NaCl, identifying the top surface characteristics that determine interactions. Finally, we demonstrate how this tool might be used in formulation design for novel antibodies via case studies.

Materials & Methods

Simulation methods

Molecular simulations were carried out according to our previously described methods.^{9,23} Briefly, molecular dynamics simulations were performed using Gromacs 5.0.5³¹ and the CHARMM36m^{32,33} force field, with the exception of force field parameters for the following excipients: sorbitol, sucrose, trehalose, proline, and Arg.HCl. Force field parameters for these excipients were taken from our previous work.^{9,23,34} Simulations were performed at pH 5.5, with amino acid protonation states set using PropKa on the PDB2PQR server.³⁵ Simulations were made charge neutral with the addition of sodium or chloride ions. All results reported here are based on simulations with a bulk excipient concentration of 0.5 m.

Energy minimization and 15 ns of equilibration were performed prior to beginning production simulations, and configurations were saved every 10 ps. Simulations were run for 70 ns, with the exception of simulations with the excipient Arg.HCl, which were run for 200 ns. The longer simulation time for Arg.HCl ensured convergence despite Arg.HCl's more complicated clustering behaviors.²² For simulations with the excipient Arg.HCl, simulations were run for at least 200 ns. The Python package **MDAnalysis**^{36,37} was used for trajectory analysis, and statistical errors were calculated using the methods of Allen and Tildesley.³⁸

Preferential interaction coefficient calculation

Overall preferential interaction coefficients were calculated from MD simulations according to the methods of Baynes and Trout³⁹ and Shukla et al.⁴⁰ For an entire protein, they can be calculated according to:

$$\Gamma_{23}(r) = \left\langle n_3(r) - n_1(r) \left(\frac{n_3^{total} - n_3(r)}{n_1^{total} - n_1(r)} \right) \right\rangle \quad (1)$$

where the subscript 1 refers to water, 2 to protein, and 3 to excipient. $n_i(r)$ refers to the number of molecules of type i within a distance r of the protein van der Waals surface. Past some distance R , usually around 8 Å, the value of $\Gamma_{23}(r > R)$ converges to a constant value. It is this value, referred to as Γ_{23} , that is compared to the experimental result.

Equation 1 can be adapted for calculating the local Γ_{23} value for each residue. When calculating local (per-residue) Γ_{23} values, the subscript 2 refers only to atoms that particular protein residue. In order to avoid double-counting, each excipient and water atom is "assigned" to its closest protein residue, and then the total number of excipient and water atoms assigned to each residue are converted back to the corresponding number of excipient and water molecules. This allows fractions of molecules to be assigned to each residue, which better describes the distribution of excipient molecules. These numbers are used to calculate the local Γ_{23} . As with the overall Γ_{23} value, we found that local Γ_{23} values tended to converge by a distance of 8 Å. Using this method, the sum of the local Γ_{23} equals the overall Γ_{23} value.

In the case of ionic excipients, the local Γ_{23} values for each excipient are calculated separately and then combined through equation 2.

$$\Gamma_{23}(r) = \frac{\Gamma_{2,+} + \Gamma_{2,-}}{2} \quad (2)$$

where $\Gamma_{2,+}$ and $\Gamma_{2,-}$ refer to the Γ_{23} values for the cation and anion, respectively, calculated

according to equation 1. Note that this slightly differs from the equation given by Record and Anderson to calculate the Γ_{23} value of the entire molecule.⁴¹ Record and Anderson subtract the net charge of the molecule from the numerator of equation 2. Thus, the overall molecule’s Γ_{23} value can be obtained by summing the local Γ_{23} values of all the residues in the molecule, then subtracting half the net charge magnitude.

Antibody dataset

The machine learning models developed here used features of individual residues to predict local Γ_{23} values. For this work, a dataset of approximately 40,000 antibody residues was developed. These residues were from the IgG1 antibodies listed in table S1, which indicates the antibody name, the PDB ID if appropriate, and the type of simulation (full antibody or Fab only). The antibodies labeled with numbers are those whose sequences were provided by AstraZeneca. Full antibody structures were generated according to the previously described methods.^{9,23,42} Additionally, the entire dataset is included in the Excel file in the supporting information. The order of the residues in this dataset is shuffled to maintain anonymity of proprietary AstraZeneca mAbs.

Each antibody residue was described by a set of 182 features, leading to a dataset of 43,816 residues x 182 features. The features used to describe each residue are an important result of this work, and are described in more detail in the results section and are summarized in Table 1. Feature values were calculated from the energy-minimized structure of the antibody in water. Prior to training the machine learning models, features were rescaled and normalized to have a mean of 0 and a standard deviation of 1.

Machine learning models

Two machine learning models were considered: elastic net (EN) models and support vector machine (SVM) models. Both were implemented using the `sklearn` Python package.⁴³ An elastic net is a simple linear model which identifies the least-squares weights for each feature

that yield the lowest error. EN models include both L1 and L2 regularization to force the weights of unimportant features to 0 and to avoid overfitting. The optimal weights, $\hat{\beta}$, are determined according to equation 3, where y refers to the target vector, X the feature matrix, and λ_1 and λ_2 are regularization parameters. The EN model is fully described by the two λ parameters.

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} (||y - X\beta||^2 + \lambda_1||\beta|| + \lambda_2||\beta||^2) \quad (3)$$

We used support vector machine models with Gaussian radial basis functions. Robert Berwick has made a clear explanation of SVM models.⁴⁴ The SVM model was implemented in Python using the `sklearn` package. The SVM model with Gaussian radial basis functions is fully described by the parameters C and γ .

Neural network models were also considered. Neural networks have been used for other antibody applications, including paratope prediction.^{25,45,46} However, the neural networks that we tested to model the local Γ_{23} values did not perform better than the SVM models (data not shown). Because neural networks have many more fitted parameters compared to SVM models, there are more opportunities to overfit the data and the resulting models would be less robust. Thus, neural networks were not included in this study.

The dataset used in developing these models consisted of the approximately 40,000 residues introduced in the previous section. These data were divided into three sets: training, validation, and testing. The testing set, used to assess the performance of the models, consisted of the data for mAbs A, B, C, and D, which account for about 10% of the total data. The data in the testing set were not included in model training. The remainder of the data made up the training and validation sets. The models were trained using 5-fold cross validation for hyperparameter tuning.

Model performance was characterized based on the root-mean-squared error (RMSE), accuracy, and true positive rate. To calculate the accuracy and true positive rate, which are classification metrics, the regression results were first binned into “low-interaction” and “high-interaction” classes, with the class boundaries chosen based on the data distribution

such that about half the data would be in each class. The "high-interaction" class was considered the "positive" class for accuracy and true positive rate calculation.

Viscosity measurements

The viscosity was measured in each excipient using a DHR-3 (TA instruments) cone and plate rheometer using a 1 degree cone with 20 mm diameter at 20°C. The dynamic viscosity was determined for shear rates between 800-2000 s⁻¹. Measurements were performed in triplicate. Solution concentrations were verified using a NanoDrop One (Thermo Scientific). The buffer conditions for all experiments were 25 mM sodium acetate, pH 5.5. For each excipient, the viscosity of mAbD at 200 mg/mL was calculated using the viscosity curve generated by measuring the viscosity at four or more concentrations between 140 and 230 mg/mL. Experimental viscosities and aggregation rates for mAbs A, B, and C, as well as the experimental methods, can be found in our previous papers.^{9,16,17,23}

Results & Discussion

Feature vector

We sought to develop a machine learning model that predicted the local Γ_{23} value over the surface of an antibody. This required subdividing the antibody surface into small units, each with a separate local Γ_{23} value. These units needed to be small enough to provide insight into the local distribution of excipient around the antibody surface, but large enough that there could be meaningful antibody-excipient interactions within each unit. For example, the entire Fv domain is not an appropriate unit. Although it is defined for any antibody and thus is transferable, it is too large to capture the local behaviors that we have previously identified to be important.⁹ Conversely, though an individual atom is also transferable, it is too small a unit to be relevant. Effectively, the unit should be on the order of the size of an excipient molecule. To fit these size criteria and to be easily applicable to any antibody, we

chose individual residues as these small units.

Any subunit of the antibody surface has artificial boundaries, as the boundary of a residue has no impact on how the protein surface in that area interacts with an excipient. Thus, the feature set needed to include information about the local neighborhood, which is its neighboring residues. This was accomplished by calculating feature values not only for atoms in the residue, but also for atoms within certain distances of any atom in the residue. For example, one feature describing a single residue is the solvent-accessible surface area (SASA) of the residue. Another feature describing that same residue is the SASA of that residue plus all atoms within 5 Å of that residue. This is an analogous idea to filtering used in image recognition or to junction tree encoding.⁴⁷

A total of 182 features were used to describe each residue. Table 1 summarizes the features used. Some of these features were also calculated over the residue and its neighboring residues, as indicated in the neighborhood column. Unless otherwise noted, neighborhood feature values were calculated at distances of 3, 5, 7, 10, 12, 15, 20, and 25 Å.

Features were split into two broad categories: sequence-based and structure-based. Sequence-based features are those that can be obtained just from the amino acid sequence, such as the residue identity. Structure-based features are those obtained from the 3D structure of the antibody. These include features such as the solvent-accessible surface area, the charge, the spatial aggregation propensity (SAP),⁸ and the spatial charge map (SCM).⁷

It is not assumed that all these features must be relevant. Instead, this feature set was developed to include any chemical property that might be relevant. The training of the machine learning models will determine the appropriate weights for each feature, with unimportant features having lower weights.

Table 1: Features used to describe each antibody residue. Category refers to the broad category for each feature (sequence-based or structural). Subtypes are listed for any related features. Features calculated both for the residue alone and for the residue plus neighboring residues within some cutoff distance (3, 5, 7, 10, 12, 15, 20, 25 Å) are identified in the neighborhood (NBH) column.

Category	Feature	Subtypes	Description	NBH?
Sequence	Hydrophobicity	-	Residue hydrophobicity	n
	Residue identity	-	One-hot vector indicating residue identity (3 encodings for histidine)	n
Structure	Protrusion index (CX) ⁴⁸	Minimum	Protrusion from mAb surface, calculated for each atom in a residue. Minimum is lowest value of any atom in the residue.	y
		Maximum	Maximum CX of any atom in the residue	y
		Average	Average CX value of all atoms in the residue	y
	Depth index (DPX) ⁴⁹	Minimum	Distance from the closest point on the mAb surface, calculated for each atom in a residue. Minimum is lowest value of any atom in the residue.	y
		Maximum	Maximum DPX of any atom in the residue	y
		Average	Average DPX value of all atoms in the residue	y

Category	Feature	Subtypes	Description	NBH?
	Electrostatic potential ²⁹	-	A charge metric modified by the distance to the surface	y
	Solvent-accessible surface area (SASA)	Total	SASA of all atoms in the residue	y
		Hydrophobic	SASA of atoms in Ala, Ile, Leu, Met, Phe, Trp, Tyr, and Val	y
		Hydrophilic	SASA of atoms in Arg, His, Lys, Asp, Glu, Ser, Thr, Asn, and Gln	y
		Positive	SASA of atoms in Arg, Hsp, and Lys	y
		Negative	SASA of atoms in Asp and Glu	y
		Backbone	SASA of backbone C, O, N, and attached H atoms	y
		Sidechain	SASA of atoms in the R group	y
	Net charge	Total	Residue net charge	y
		Exposed	Sum of partial charges of atoms with SASA > 0	y
	Shape index ²⁹	-	Description of concavity/convexity	^a
	SAP ⁸	-	Measure of surface-exposed hydrophobicity of atoms within 5 Å	Only 5 Å ^b
	SCM ⁷	-	Measure of exposed charge within 10 Å	Only 10 Å ^c

Category	Feature	Subtypes	Description	NBH?
	Secondary structure	-	One-hot vector indicating secondary structure (α -helix, β -sheet, loop)	n
	Fractional exposure	-	Ratio of the residue’s SASA to the standard exposure of the residue in Ala-X-Ala	n

^a The shape index was not calculated for the residue alone and neighborhood values were calculated at 3, 4, 5, 6, 7, 8, 9, and 10 Å.

^b The SAP tool was designed for calculations at 5 Å, as discussed by Chennamsetty et al.⁸

^c The SCM tool was designed for calculations at 10 Å, as discussed by Agrawal et al.⁷

Antibody dataset

The data set used in this study consisted of the residues of the 41 antibodies listed in table S1. Some structures were for full antibodies, while others were just for the Fab domains. We have found that the local Γ_{23} values residues in an antibody in a Fab-only simulation generally match those of the corresponding residues in a full antibody simulation, as the interactions we are examining are local. Thus, we could mix data from Fab-only and full antibody simulations. Only residues with nonzero surface exposure were included in the data set. This led to a data set of approximately 40,000 residues, which was randomly divided into training and testing sets.

In principle, this approach should be applicable to any protein. However, the data set contained only residues from antibody structures, as that was the focus for this work. If this approach is to be extended to include other types of proteins, similar proteins should be included in the training data.

The target values are the local Γ_{23} values for each residue, calculated as described in the

methods section. The feature values were calculated from energy minimized structures. We compared the performance of models instead trained with features averaged over a 50 ns simulation of the mAbs in water and found that the RMSEs and accuracies were virtually identical. Because obtaining energy minimized structures is much less time consuming and there was no loss in performance, we chose to use features calculated from energy minimized structures. Figure S1 also demonstrates visually that both these approaches lead to similar performance.

Machine learning models

Two types of machine learning models were considered: elastic nets and support vector machines. Separate models were developed for each excipient included in this study: sorbitol, sucrose, trehalose, proline, Arg.HCl, and NaCl. Generally, the same hyperparameters were used for the carbohydrates, proline, and NaCl, while a different set of hyperparameters were used for Arg.HCl. The hyperparameters for the EN and SVM models developed are shown in table 2. A visual example of the performance of the NaCl models on mAbC, an antibody excluded from the training and testing set, is shown in figure 1.

Table 2: Hyperparameters for machine learning models. Carbohydrates refers to sorbitol, sucrose, and trehalose.

	EN		SVM	
	λ_1	λ_2	C	γ
Carbohydrates, proline, NaCl	0.01	0.0001	40	0.001
Arg.HCl	0.000316	4.64×10^{-7}	55	0.001259

Figure 1 shows how, while neither model is a perfect match for the simulation result, both capture much of the relative excipient distribution around the antibody. The root-mean-squared error (RMSE), accuracy, and true positive rates (TPR) are reported in table 3. The SVM models reduce error and improve accuracy as compared to the naïve baseline case, which predicts the average Γ_{23} value for all residues. In addition, the SVM models have

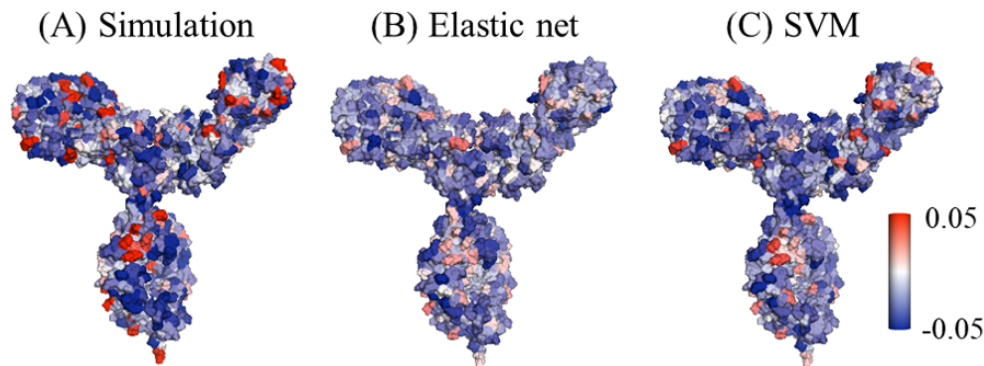


Figure 1: Views of mAbC interactions with NaCl, colored by the local Γ_{23} value (based on a bulk excipient concentration of 0.5 m). (A) shows the simulation result, which is the target of the machine learning models. (B) and (C) show the elastic net model and SVM model, respectively. Both clearly capture the trends in the distribution of NaCl around the antibody, with the SVM model performing slightly better. However, neither perfectly captures the magnitude of all the interactions.

significantly improved accuracies and TPRs over the elastic net models.

Table 3: Root-mean-square errors, accuracies, and true positive rates for the naïve baseline model, elastic net, and SVM models for each excipient. The naïve baseline predicts the average Γ_{23} value for each residue. Results shown for performances on the test set. For all excipients, the SVM models have slightly lower error and higher accuracies compared to the elastic net models.

Excipient	Naïve baseline			EN			SVM		
	RMSE	Acc.	TPR	RMSE	Acc.	TPR	RMSE	Acc.	TPR
Sorbitol	0.032	0.45	0	0.017	0.85	0.92	0.017	0.85	0.90
Sucrose	0.041	0.45	0	0.023	0.85	0.87	0.021	0.86	0.89
Trehalose	0.038	0.45	0	0.021	0.85	0.88	0.019	0.86	0.89
Proline	0.024	0.54	0	0.017	0.81	0.89	0.016	0.83	0.86
Arg.HCl	0.092	0.57	0	0.086	0.65	0.61	0.083	0.78	0.80
NaCl	0.025	0.50	0	0.019	0.79	0.84	0.017	0.82	0.85

Elastic net feature weights

Often, it can be challenging to gain biological insight from statistical learning algorithms.⁵⁰

One benefit of the elastic net, which is a linear model, is that it provides some insight into the features that are important to the model. A trained elastic net is a set of weights,

one for each feature, with the weighted sum of the feature vector yielding the target value. Prior to training the elastic net, all features are rescaled and normalized, meaning that the values for each feature generally have an average of 0 and a standard deviation of 1. This normalization allows the weights to be compared across different features, as the possible magnitudes of each feature are approximately the same. Figure 2 shows, for each excipient, the features with weights of at least 4%. Note that figure 2 shows weights with positive or negative values. Features with a positive weight contribute positively to the local Γ_{23} value, meaning higher values of that feature correspond to more local excipient molecules, while features with a negative weight contribute negatively, meaning higher values of that feature correspond to more local water molecules.

Interestingly, the hydrophilic SASA is the only feature with a contribution of at least 4% to the local Γ_{23} values for each type of excipient. For the carbohydrates and proline, it has a strong negative impact, while for NaCl it has a weak negative impact and for Arg.HCl it has a slightly positive impact. This suggests that the carbohydrates and proline tend to interact less with hydrophilic residues compared to Arg.HCl and NaCl, which makes sense given that Arg.HCl and NaCl are ionic compounds.

As shown in figure 2, sorbitol, sucrose, and trehalose had similar high-weight features. These included the hydrophobic surface area, which had the highest positive weight for all three excipients, as well as the positive, negative, hydrophilic, and sidechain surface areas, which had some of the highest negative weights. These similarities suggest that the three excipients interact through a similar mechanism, which makes sense, as they are all carbohydrates and expected to interact primarily through the alcohol groups. The positive contribution of the hydrophobic surface area agrees with our previous result indicating that the carbohydrate excipients tend to interact more strongly with hydrophobic regions.⁹ The negative contributions of the hydrophilic and charged surface areas to the local Γ_{23} indicate that these hydrophilic residues tend to interact with water rather than the carbohydrate excipients.

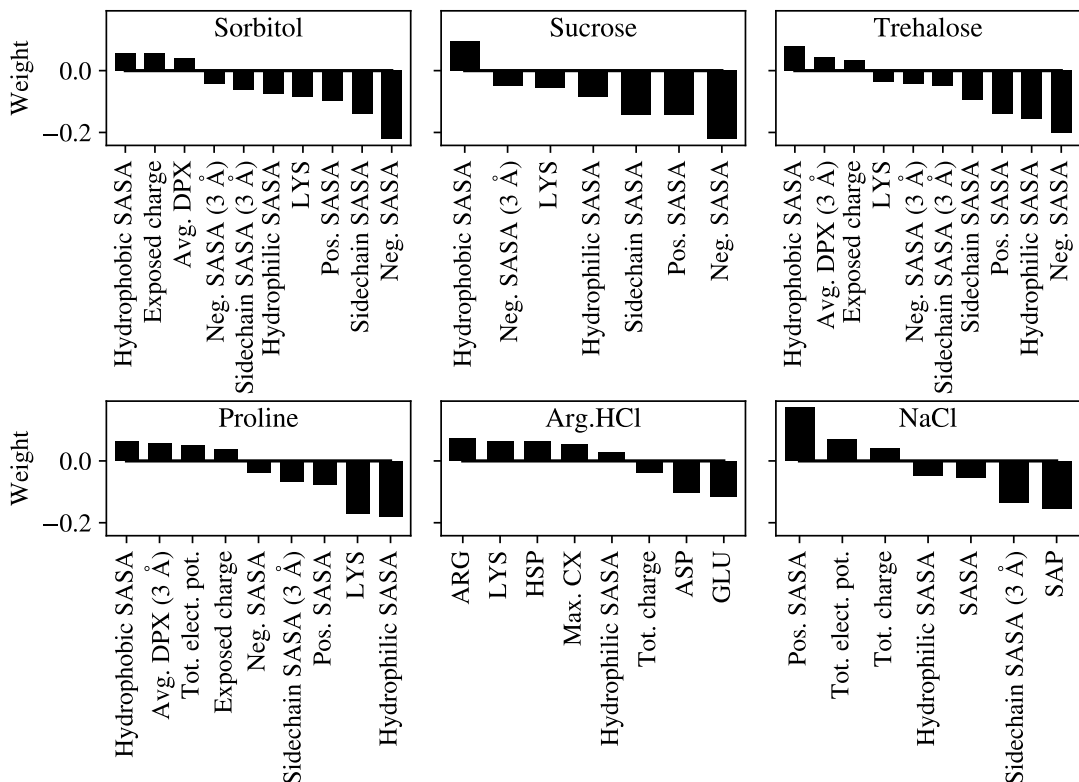


Figure 2: Plots of the feature weights from the elastic nets for features with weights above 4%. Positive values indicate a positive contribution to the local Γ_{23} value, and negative values indicate a negative contribution. Feature explanations can be found in table 1. In general, similar features show up as the top features for sorbitol, sucrose, and trehalose, indicating that the underlying mechanism of interaction of these excipients with the antibody is the same. Interestingly, there is some overlap with the top features for proline interactions, indicating some similarities in how these osmolytes interact. For Arg.HCl, the identity of the surface residue tends to have a high impact, as indicated by many of the top features being a particular type of residue. For NaCl, charge features tend to be most important.

Interestingly, many of the top features for proline also appear among the top features for the carbohydrate excipients. For example, the hydrophobic SASA has the highest positive weight, and the positive and hydrophilic SASAs have some of the most negative weights. As with the carbohydrates, this suggests that proline interacts more strongly with hydrophobic residues, while water interacts more strongly with hydrophilic residues. Previous studies have shown that the pyrrolidine side chain of proline interacts with hydrophobic residues via CH- π interactions.^{51,52}

For Arg.HCl, many of the top features are residue names. These features are from

the one-hot residue name feature set, in which the one high value indicates the residue identity. Specifically, all five charged residues appear in the top five features for Arg.HCl. The three positively charged residues have positive weights, while the two negative residues have negative weights. However, the net charge of the residue has a negative weight, leading to a positive contribution of the negatively charged residues to the local Γ_{23} values. Because the local Γ_{23} value is based on all the features in the feature set, there can be several competing effects. The Arg.HCl plot does show that features related to charge are important and may provide further insight into these competing effects.

For NaCl, the top features are mostly related to charge or hydrophobicity. It makes sense that the feature with the highest positive weight is the positive charge SASA, since NaCl is expected to interact primarily through electrostatic interactions. The feature with the most negative weight, the SAP, is effectively a measure of surface hydrophobicity. This suggests that more hydrophobic regions on the surface interact more strongly with water than with NaCl.

Case studies for formulation design

Previous studies^{9,23} have identified important antibody-excipient interactions that are thought to be related to the observed experimental behavior. Here, we examine if those interactions are captured by the SVM model and describe how that insight might be used in formulation design. Then, we demonstrate how we applied this approach to an antibody prior to assessing its viscosity experimentally. In all the case studies presented here, the antibodies were excluded from the training and testing sets. Thus, the SVM results presented represent the model performance on an antibody entirely not previously encountered by the model.

Aggregation

Our previous studies on mAbA indicated that, at excipient concentrations of 100 mM, Arg.HCl and NaCl increased the aggregation rate by factors of about 1.6 and 2.3, respec-

tively, while proline had no effect.²³ At 200 mM, the carbohydrates reduced aggregation.⁹ Because only the ionic excipients increased aggregation, these increases are thought to be related to charge effects. For mAbA, these behaviors were examined by considering the net local Γ_{23} values of the Fv domain as a whole, as well as of the positive and negatively charged residues within the Fv domain. Figure 3 compares these values as calculated from the simulations (A) and as calculated based on the SVM models (B).

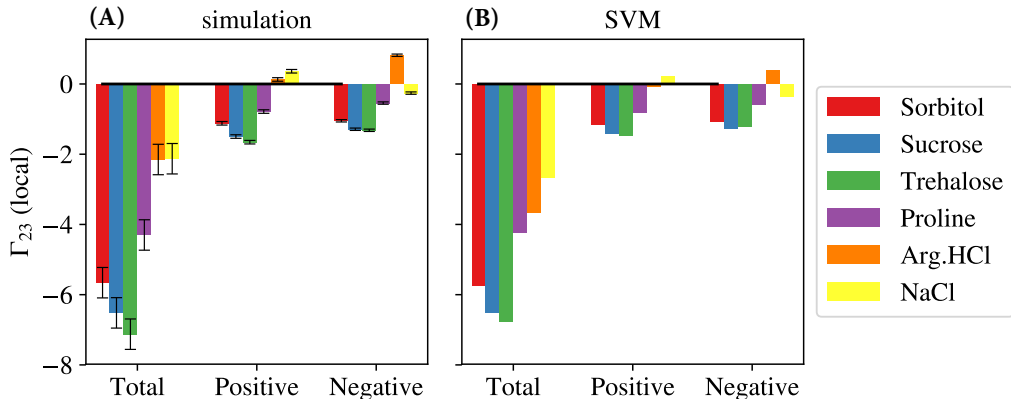


Figure 3: Sums of the Γ_{23} values in the variable region of mAbA from (A) simulations with each excipient or (B) the SVM model outputs, based on a bulk excipient concentration of 0.5 m. The results are further broken down by summing over just the positively or negatively charged residues in the Fv. The SVM models generally capture the appropriate magnitudes of the total Γ_{23} values, although the SVM model for Arg.HCl tends to predict more exclusion. The SVM model does capture that Arg.HCl is included near negative residues, while NaCl is excluded. This behavior may be related to why NaCl increases mAbA aggregation more than Arg.HCl.

Near positive residues, the simulations (figure 3A) indicate slight inclusion of Arg.HCl and NaCl, while near negative residues, they indicate inclusion of Arg.HCl but exclusion of NaCl. As shown in figure 3B, the SVM models capture all these behaviors within the error tolerance. Our previous study suggested that the increase in aggregation of mAbA in Arg.HCl and NaCl was likely due to a reduction in electrostatic repulsion in the presence of these excipients, as mAbA has strong positive charges on the Fv and Fc at pH 5.5.²³ The weaker interaction of NaCl with negative residues may contribute to the greater increase in aggregation rate in the presence of NaCl (increase of 138%) compared to Arg.HCl (increase

of 57%), and this result is captured by the SVM models.

We performed similar analysis to understand the impact of trehalose on the aggregation of mAbB. Our previous study found that trehalose reduced the monomer loss rate of mAbB by about 32%,¹⁷ and analysis of local excipient interactions suggested that this was due to interactions of trehalose with hydrophobic residues.⁹ We confirmed that this behavior was captured by the SVM models. Both the simulation and the SVM model indicate that sum of the local Γ_{23} value for trehalose near hydrophobic residues in the Fv was -0.8. Additionally, both indicate the sum of the local Γ_{23} values near aromatic residues in the Fv was -0.3. In comparison, if trehalose was excluded the average amount from these types of residues, the local exclusions would be -3.3 and -2.8, respectively. This result, captured by the SVM models, indicates that trehalose is much less excluded than average from hydrophobic and aromatic residues in the Fv, and these interactions could lead to the observed disruption of aggregation.

Viscosity

Our previous study on mAbC found that its viscosity at 100 mg/mL was reduced from about 35 cP at pH 5.5 to about 5 cP with the addition of 100 mM Arg.HCl or NaCl.²³ The high viscosity of mAbC is likely related to the negative charge patches in the Fv, which may also contribute to the propensity for reversible self-association.⁵³ These negative charge patches are shown in red in figure 4A. Figure 4B-C show the local Γ_{23} values for Arg.HCl and NaCl calculated from simulation, while D-E show the local Γ_{23} values output by the SVM models. While the simulation and SVM results are not identical, the SVM results do capture the important excipient interactions with the circled high-SCM region that are expected to be responsible for the observed reduction in viscosity.

The simulation results indicate net inclusion of Arg.HCl (local $\Gamma_{23} = 2.9$) and NaCl (local $\Gamma_{23} = 0.2$) with the highest-SCM region circled in figure 4A. The strong interactions with this negatively charged region are expected to disrupt mAbC-mAbC self-interactions that

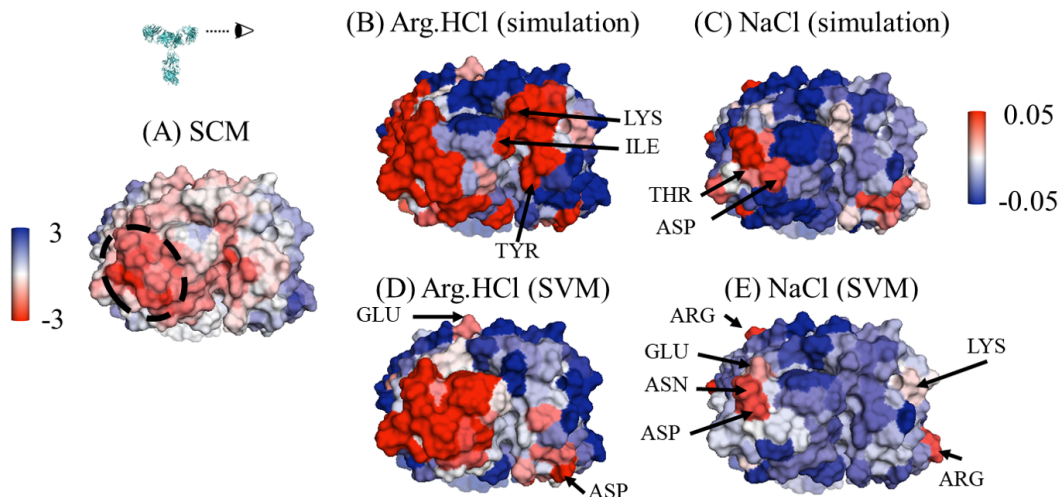


Figure 4: Views of the mAbC variable region. (A) shows the Fv colored by the SCM score, with red indicating exposed negative charge (left colorbar). (B)-(E) show the Fv colored by the local Γ_{23} value (right colorbar). (B) and (C) are results from simulations with the excipients Arg.HCl or NaCl, respectively, while (D) and (E) are the predicted values using the SVM models, all based on a bulk excipient concentration of 0.5 m. Both the SVM model and the simulation result indicate that these excipients are net-included near the lowest-SCM region circled in (A).

lead to increased viscosity. The SVM results also indicate net inclusion of Arg.HCl (1.6) and NaCl (0.02), though both are lower in magnitude than the corresponding simulation results. Even so, the SVM models still capture the much stronger than average interactions in the negatively charged region, which is necessary for implementing the SVM models in formulation design. For example, if a proposed antibody were found to have a large amount of negative charge in the Fv domain, like mAbC, the SVM model could be used to check if ionic excipients will interact strongly in that region. If so, the results with mAbC suggest that the viscosity in those excipients will be reduced to acceptable levels.

Demonstration of approach: mAbD viscosity

For the previous case studies, the experimental work was done prior to the simulation work. In order to perform an evaluation on an antibody which we had not previously studied, we evaluated a new antibody, mAbD, first through simulation and then through experiment.

We began by using the SAP⁸ and SCM⁷ tools to characterize the surface of the Fv domain. The low overall SCM score, 375 ± 22 , and the high net positive charge overall, +36, and on the Fv, +9, suggested that the antibody would have low viscosity at 100 mg/mL. There is a small region of exposed negative charge in the Fv domain, but mostly exposed positive charge, which is expected to lead to electrostatic repulsion and thus low viscosity. The low SCM score and the relatively small amount of exposed negative charge in the variable region suggest that charge effects do not dominate the viscosity behavior of mAbD.

Based on this analysis of the mAb Fv surface and our experience with mAbs A, B, and C, we expected carbohydrates to have little effect on the viscosity or to increase the viscosity. He et al. found that sugars tended to increase the viscosity of antibodies at high concentrations.⁵⁴ This generally agreed with our previous results for the viscosity behavior of mAbs B and C in carbohydrates,⁹ although there were some exceptions based on specific self-association behaviors.

We expected slightly more complicated behaviors in the presence of proline, Arg.HCl, and NaCl. From our previous work with these excipients,²³ we knew that they can impact viscosity through interactions with charged residues and hydrophobic residues. Figure 5 shows the local interaction maps of these excipients with the variable region of mAbD. In addition, it shows the SCM scores mapped onto the variable region, with the two positively charged regions circled.

Similarly to mAbA, we expected that interactions with these positively charged regions might disrupt the electrostatic repulsion between molecules, leading to higher viscosity. According to the SVM models of the mAbD behavior in 0.5 m excipient, proline and Arg.HCl are fairly strongly excluded from these regions, with local Γ_{23} values of -0.51 and -1.04, respectively. Importantly, these values are both more negative than would be expected if the excipients were excluded the average amount from these residues. In general, exclusion from this region suggests that the electrostatic repulsion between mAbD molecules will be maintained, leading to the same or slightly lower viscosity than in buffer.

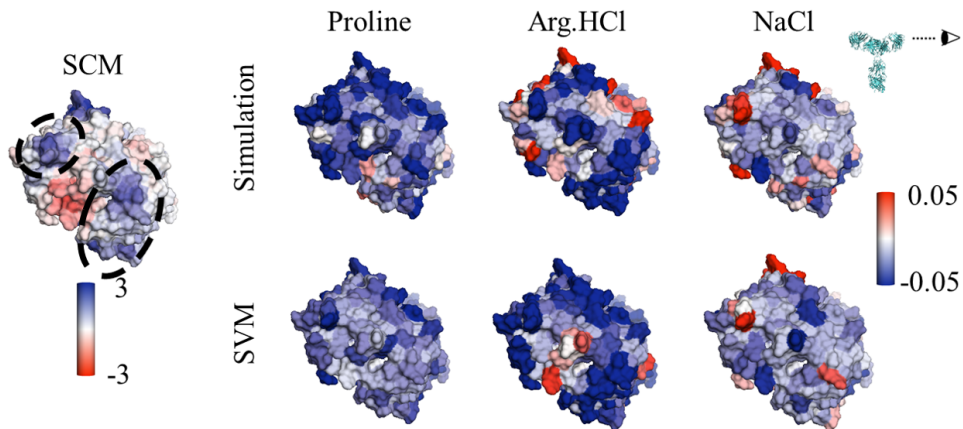


Figure 5: Views of the mAbD variable region colored by the SCM score (far left) or by the local Γ_{23} values with each excipient. The top row of plots is colored by the local Γ_{23} values based on simulations performed with bulk excipient concentrations of 0.5 m, while the bottom row of plots is colored by the local Γ_{23} values predicted by the SVM model. Both the simulation and SVM models indicate that proline and Arg.HCl are excluded from the circled positive charge regions, while the net Γ_{23} value for NaCl near these regions is around 0. This suggests that NaCl interferes more with the electrostatic repulsion of mAbD from other mAbD molecules at pH 5.5, leading to the observed increase in viscosity compared to behavior in the buffer.

The SVM model indicates slight exclusion of NaCl from this region, with a local Γ_{23} value of -0.08. However, this value is higher than what would be expected if NaCl was excluded from each residue in this patch by the average local Γ_{23} value. This average exclusion would correspond to a local Γ_{23} value of -0.29. Thus, NaCl has above-average interactions with this positively charged region of mAbD. This above-average interaction was expected to correspond to a slight increase in the viscosity of mAbD in NaCl, because it could disrupt the electrostatic repulsion between mAbD molecules.

We compared the SVM results with the simulation results. The simulation results also indicated strong exclusion of proline ($\Gamma_{23} = -0.7$) and Arg.HCl ($\Gamma_{23} = -1.2$) from the circled high-SCM regions. The simulation indicated slight inclusion ($\Gamma_{23} = 0.1$) of NaCl with these regions. This agrees with the SVM results which found that proline and Arg.HCl are more excluded than average, while NaCl is more included than average.

Finally, we experimentally validated these results by measuring the viscosity of mAbD

in buffer and in the presence of each excipient. As we expected, mAbD had a low viscosity, around 10 cP, at 100 mg/mL in 25 mM sodium acetate buffer, pH 5.5. In order to observe and differentiate between the impacts of the different excipients, we measured the viscosities at higher concentrations of antibody. Figure 6 shows the viscosity behavior in each excipient at 200 mg/mL.

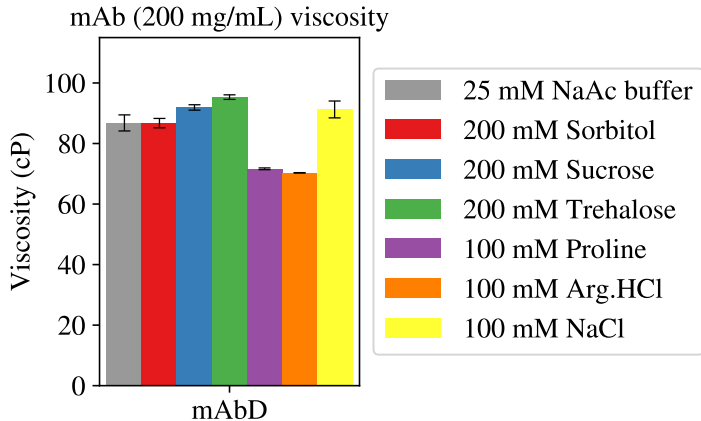


Figure 6: mAbD viscosity at 200 mg/mL at pH 5.5. Sucrose, trehalose, and NaCl increase the viscosity, compared to the viscosity in the buffer, while proline and Arg.HCl reduce the viscosity.

Figure 6 confirms our predictions for the viscosity behavior of mAbD in each of the excipients. Of the three carbohydrates, sorbitol has no effect on the viscosity while sucrose and trehalose increase the viscosity. Proline and Arg.HCl reduce the viscosity, while NaCl increases the viscosity. This demonstrates that the SVM models were able to capture the key interactions that might be responsible for the observed viscosity behavior. It is important to note that, based on our experience, it is necessary to observe the properties of the mAb surface, such as SAP and SCM, to understand the potential impact of each excipient. The surface characteristics will influence the underlying aggregation and viscosity mechanisms, which in turn are important for understanding how these behaviors will change in the presence of excipient.

Formulation design scheme

Based on these studies of antibody-excipient interactions, we propose the following formulation design scheme, also described in figure 7:

1. Candidate mAb sequence is converted to a structure using homology modeling
2. Surface characteristics, such as SAP and SCM, are modeled to identify problem patches which might contribute to elevated aggregation or viscosity
3. The SVM models for local Γ_{23} values presented here are used to predict local interactions with excipients
4. The local Γ_{23} sums near potentially problematic patches, such as those with exposed positive or negative charge, are used to predict the effect of the excipient on the mAb's stability
5. If there are expected to be aggregation or viscosity behaviors not addressed by the addition of excipients, point mutations can be performed and this process repeated from step 1
6. If aggregation and viscosity behaviors are expected to be addressed in the presence of certain excipients, the mAb structure and the stabilizing excipients move forward in the design and development process

This process can be applied to any new candidate antibody. Step 4 describes the most complicated stage, which involves analyzing the local excipient interactions in light of the antibody surface characteristics to predict the effect. Each of the case studies presented previously shows an example of this analysis being carried out and can be used as a guide in performing this analysis. The final outputs of this analysis are an optimized mAb sequence as well as the excipients expected to have a benefit in terms of propensity for aggregation and viscosity behavior.

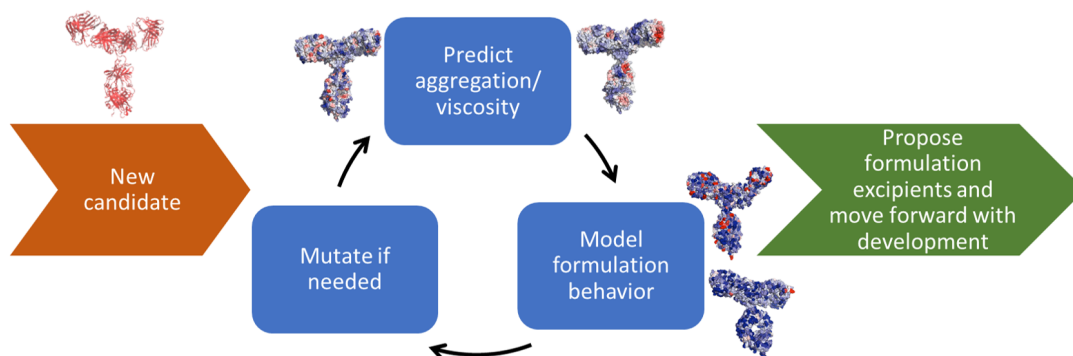


Figure 7: Proposed computational formulation design workflow.

Conclusions

In conclusion, we have developed a feature set that can be used to numerically represent the characteristics of small regions of an antibody’s surface and demonstrated how these features can be used to model local antibody-excipient preferential interaction coefficients. The feature set contains both sequence information and structure information, such as charge, surface area, and hydrophobicity. We demonstrated that these features could be calculated for the energy-minimized antibody structure and used to model the local Γ_{23} values of sorbitol, sucrose, trehalose, proline, Arg.HCl, and NaCl with up to 86% accuracy using SVM models. In case studies with mAbs A, B, and C, the results of these models capture the relative strengths and weaknesses of the interactions of these excipients with regions of the mAb surface that are thought to be related to aggregation and viscosity behavior. This approach was validated by applying it to predict the impact of these excipients on the viscosity of a new antibody, mAbD, prior to carrying out experimental work. Finally, we presented a framework for computational formulation design using machine learning SVM models that have the potential to predict local antibody-excipient interactions to enable a mechanistic approach to directional formulation design.

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